Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C++, C, SQL, JavaScript

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Backend & Software Development: FastAPI, Flask, Node.js, REST APIs, API Development, Middleware, Multithreading, Authentication, React.js

Databases & Systems: MySQL, SQLite, MongoDB, Redis, In-Memory Data Structures, Caching, Concurrent Programming

Software Engineering Practices: Git, GitHub, PyTest, TTD, Debugging, Unit Testing, Code Quality, Performance Optimization

Tools & Deployment: Docker, Render, Vercel

EXPERIENCE

VOIS for Tech University Engagement Program **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyze, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimized data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

PROJECTS

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

API Rate Limiter Library **|LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

Docwise **|LINK**

- Built a RAG-based document Q&A system using Google Gemini, NLP, TF-IDF retrieval, and intelligent chunking, achieving 95%+ query accuracy on PDFs exceeding 200 pages.
- Designed an end-to-end document processing and retrieval pipeline for text extraction, chunking, vectorization, context retrieval, and LLM-based response generation.
- Optimized retrieval quality through prompt engineering, relevance tuning, and hallucination-aware validation, improving response reliability and contextual accuracy.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**